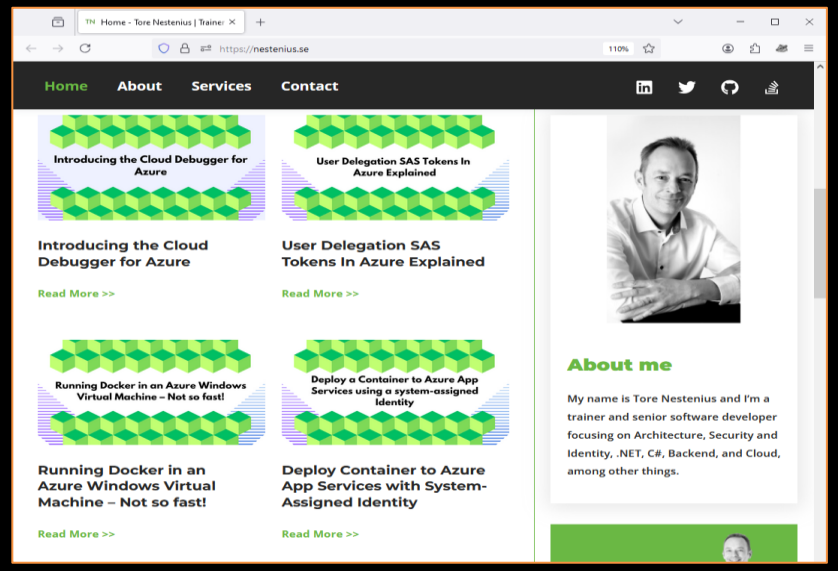
How Does a Coding Agent Work?

nestenius.se

Microsoft Most Valuable Professional

About Tore Nestenius

Work https://tn-data.se



Blog https://nestenius.se

.NET Skåne

https://meetup.com/net-skane

Have a question about this session or beyond?

tore@tn-data.se

First, One Important Thing

Do Ask Questions!

The AI Revolution

The AI Revolution

Claude skapar kaos – konsultfirma slutar ta betalt för kodning: “Oundvikligt”

WIZARDWORKS SAAS AI 28 jan 2026, kl 11:26

https://www.breakit.se/artikel/45281/claude-skapar-kaos-konsultfirma-slutar-ta-betalt-for-kodning-oundvikligt

The AI Revolution

Rebel and Bird
1,882 followers
Visit website
3d •

Är konsulttimmen död?
Vi tror det. Den traditionella modellen där man säljer tid istället för värde har nått vägs ände.
Därför genomför vi på Rebel and Bird nu vår största transformation någonsin. Vi ställer om hela verksamheten till Agentic First.
Det innebär att vi lämnar löpande räkning bakom oss och lanserar Output-As-A-Service. Genom att kombinera våra seniora konsulter med autonoma AI-agenter kan vi nu erbjuda en ekvation som tidigare var omöjlig:
-Svensk spetskompetens och fysisk närvaro.
-Korta ledtider och tät dialog.
-En prislapp som matchar offshore-leverans. Till en fast månadskostnad.

Varför välja mellan kvalitet och pris när tekniken nu tillåter båda?
Vi bygger framtiden istället för att vänta på den. Är ni redo att sluta köpa timmar och börja prenumerera på resultat?

The AI Revolution

Peter Scheffer • 2nd
AI implementations that deliver real, measurable ROI for busine...
3d •

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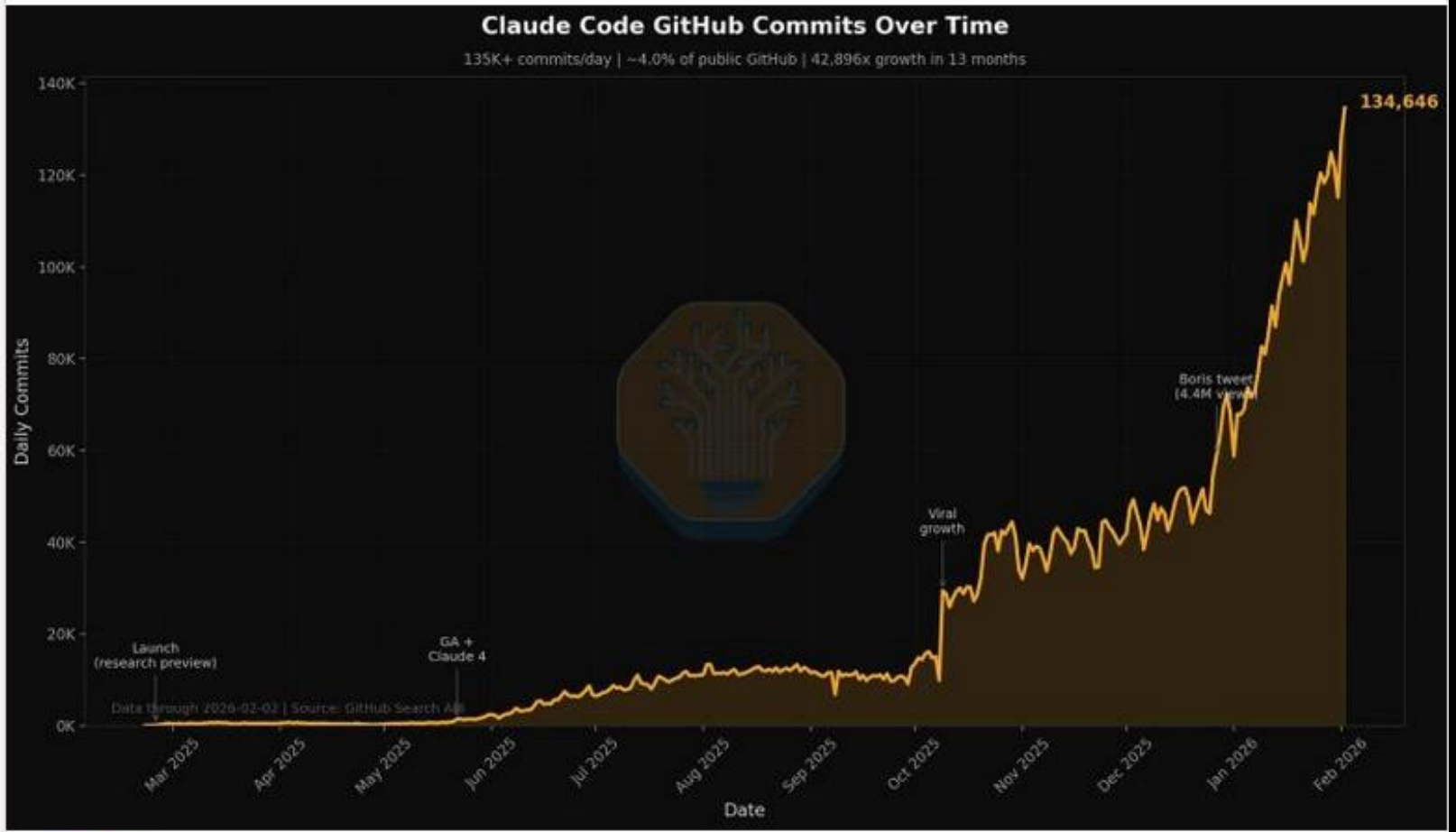
"4% of GitHub public commits are being authored by Claude Code right now. At the current trajectory, we believe that Claude Code will be 20%+ of all daily commits by the end of 2026. While you blinked, AI consumed all of software development."



https://nestenius.se

How many of you…

Claude Code,
Copilot, Cursor…

Have used a **Coding Agent**?

Could **explain** how it actually works?

Why Did I Write This Talk?

Why Did I Write This Talk?

I want to teach Agentic AI Development

But how should I teach it?

Why Did I Write This Talk?

Just writing prompts is not enough!

```
Create a Hello World app
```

```
...
```

```
Create a C compiler that can
compile the Linux kernel!
```

Coding Agent

This will not teach you agentic coding!

What do we need instead?

What do we need instead?

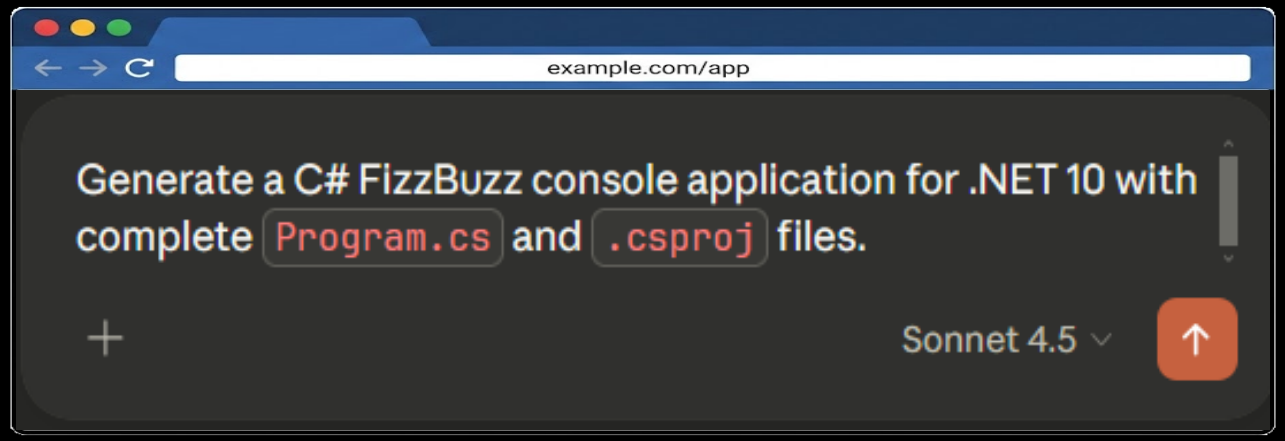
We need Mechanical Sympathy!

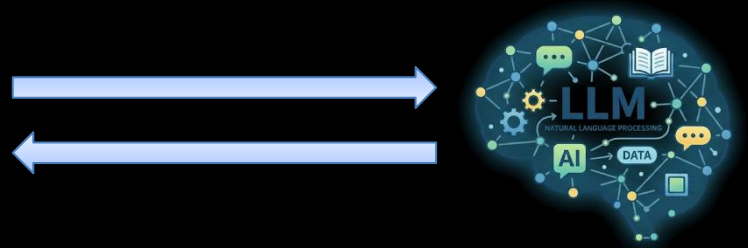
- Understand what happens **under the hood**
- Know how the agent **talks** to the LLM
- Understand **tools** and how the agent uses them
- Learn what drives token **usage** and **cost**

You don't drive a race car by ignoring the engine

What Is Agentic Coding?

Most of us started by using LLMs in the browser





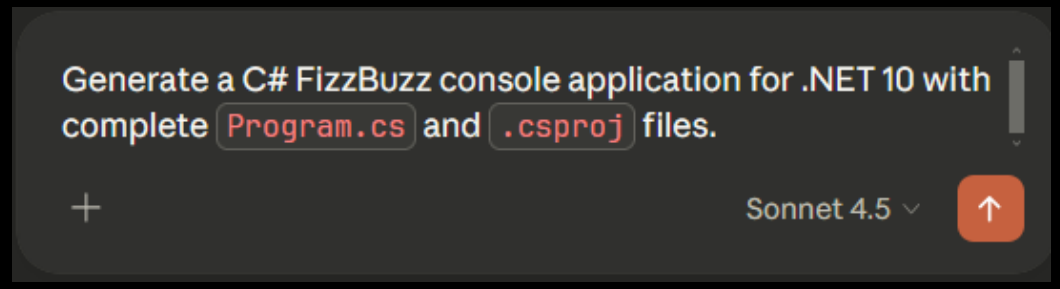
Why doesn’t this scale to real software work?

The developer becomes the message transport

Source Code

Copy / Paste

Web Browser

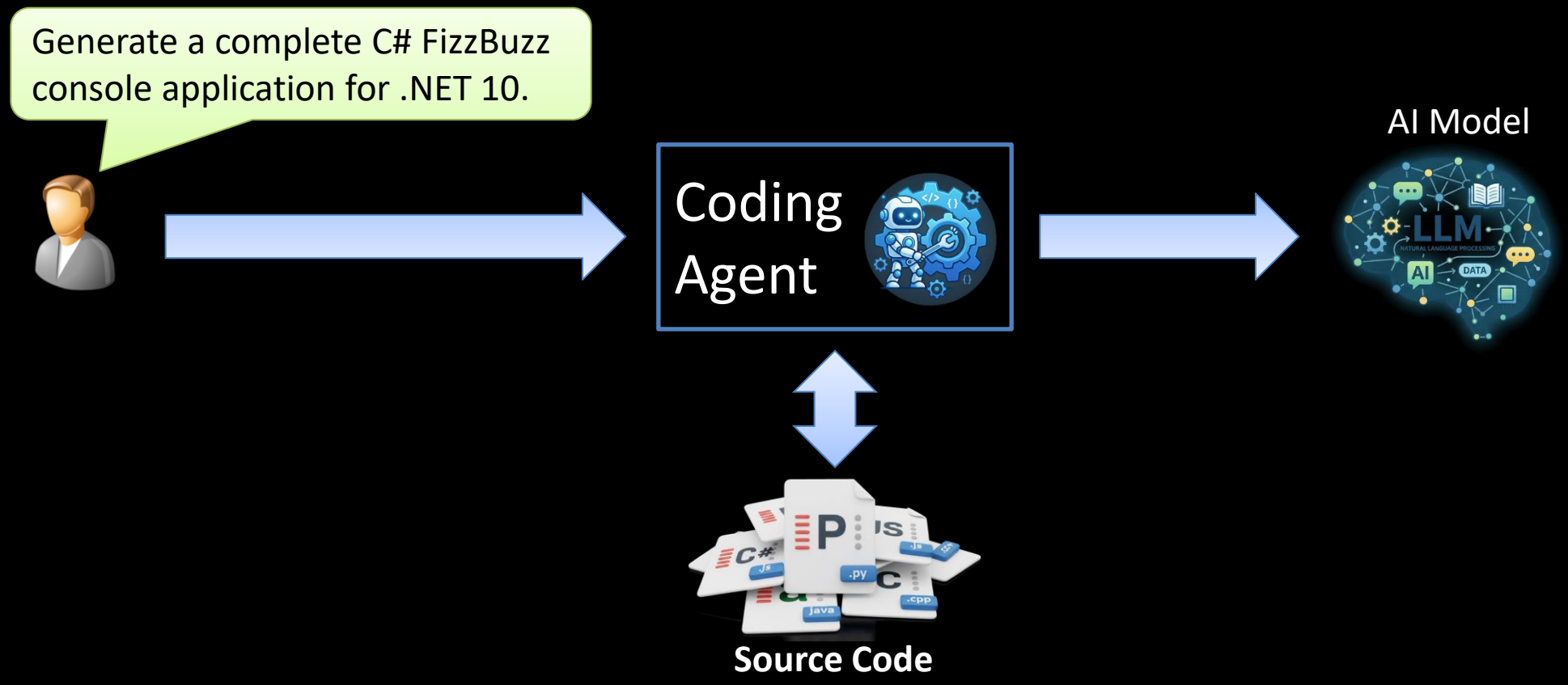


Model

Copy -> Paste -> Error -> Copy -> Paste -> Repeat...

Can we remove the human from the feedback loop?

Agentic Coding: the AI operates directly on your codebase



So, how does this work?

Coding Agents

Meet your new coding partner: The Coding Agent

How can it access my source code files?

Large Language Models (LLMs) operate only on text

Coding Agent

```
what is the problem with the code in my
Program.cs file?
```

```
I can't see the code yet. Please paste
the contents of your Program.cs file
```

How can we observe this interaction?

Demo #1

c:\AgentDemo

Claude Code

HTTP Inspector

Table view of all proxied requests with headers, bodies, SSE events, and timing details.

Conversation View

Chat-style display of Claude conversations showing messages, tool use, and responses.

How do we give it access to the source code?

Source Code

We can add a set of **tools** to our coding agent

- File I/O
- Run commands
- Web fetch
- Web search
- …

Coding Agent

How can the LLM use these tools?

The coding agent describes the available tools to the LLM

Program.cs

Read file
Program.cs

Coding Agent

File Tools

```
what is the problem with the code in my
Program.cs file?

If you want to read a file, respond with
"Readfile: filename"
```

```
Readfile: Program.cs
```

```
Program.cs:
Console.WriteLine("Hello, World!);
```

```
The problem is a syntax error caused by
an unterminated string literal.
```

All coding agents work in a similar way!

Demo #2 – Exploring the tools

```
Create a simple FizzBuzz console application in C#
targeting .NET 10.

- Only create Program.cs and the .csproj file.
- Assume Windows environment.
- Do not use bash or execute shell commands.
```

Claude Code

Model Context Protocol (MCP)

First, we had the web, for humans to consume

HTML/CSS/JS

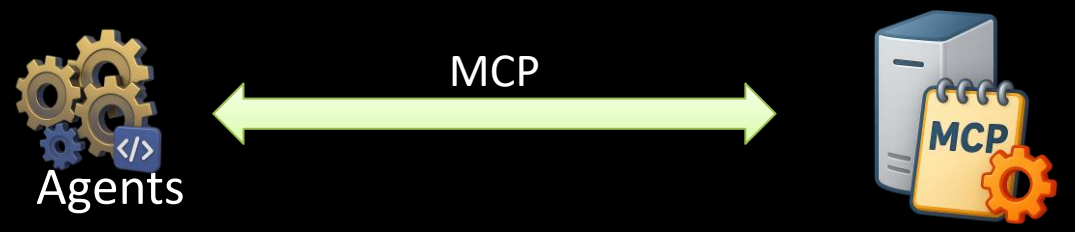
Then came the need for APIs

REST / gRPC

If we already have APIs, why do we need MCP?

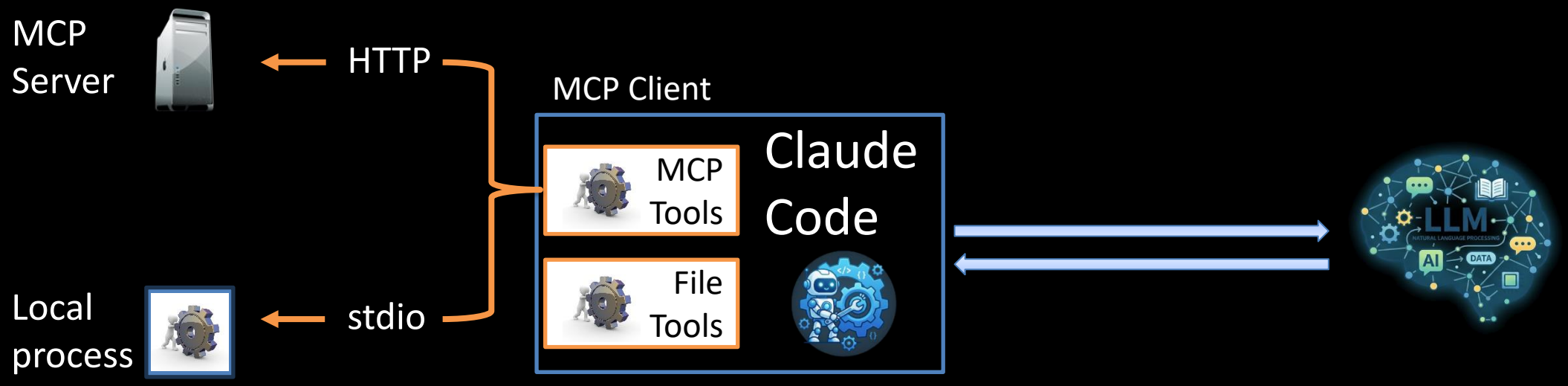
APIs are designed for applications, not agents

MCP lets you expose services and tools to agents



What does MCP specify?

MCP specifies the protocol between an agent and server



How do we use these MCP tools in our coding agent?

MCP tools are exposed to the model as regular tools

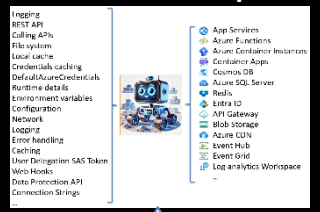
HTTP

```
What is the CloudDebugger?

fetch_CloudDebugger_documentation
Fetch entire documentation file from GitHub
repository: tndata/CloudDebugger.

Useful for general questions. Always call this
tool first if asked about tndata/CloudDebugger.

search_CloudDebugger_docs ...
search_CloudDebugger_code ...
fetch_generic_url_content ...
...
```

The model never communicates directly with MCP servers

Demo #3 – Exploring MCP

GitHub Repo



https://gitmcp.io/tndata/CloudDebugger

What are the main features of the CloudDebugger?

MCP Tools

Claude Code

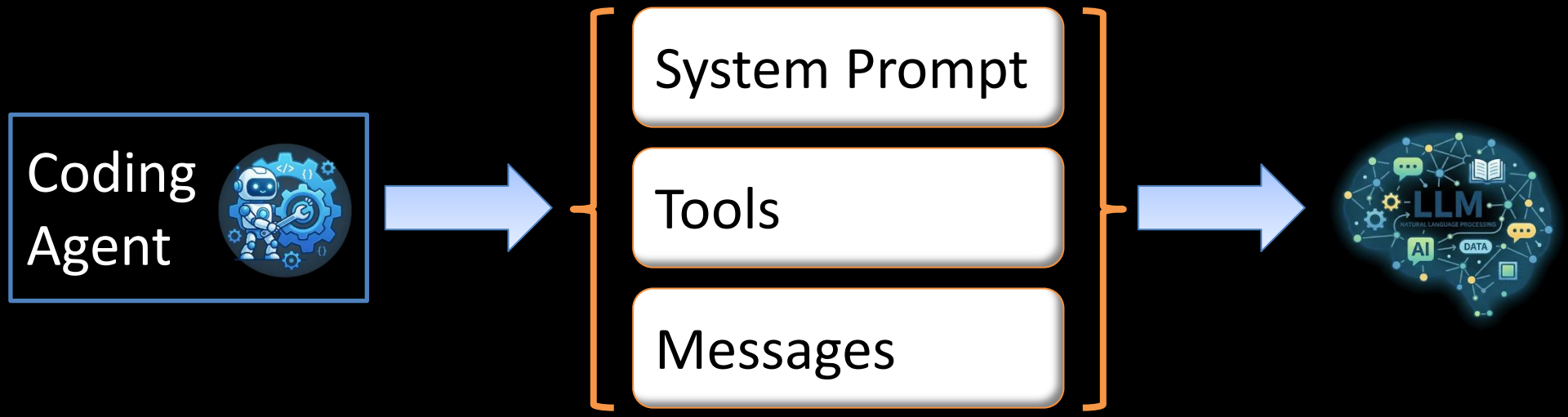
```
claude mcp add --transport http clouddebugger https://gitmcp.io/tndata/CloudDebugger
```

Register MCP server

The Three Pieces of an LLM Request

The Three Pieces of an LLM Request

Every LLM request consists of three parts:



What are the differences?

System Prompt: The LLM's instruction manual

Coding Agent

```
You are an expert C# coding assistant.

Your role is to help developers write, debug, and
improve C# code.

Guidelines:
- Use modern C# features and best practices
-  Follow SOLID principles
-  Write clean, maintainable code
-  Explain your reasoning when making changes
-  Always verify your solutions work correctly


Be concise, helpful, and focus on producing working
code.
```

Included in every request

Tools: The list of available tools that the LLM can call

Coding Agent

```
Task
TaskOutput
Bash
Glob
Grep
ExitPlanMode
Read
Edit
Write
NotebookEdit
WebFetch
WebSearch
TaskStop
AskUserQuestion
Skill
EnterPlanMode
TaskCreate
TaskGet
TaskUpdate
TaskList
ToolSearch
```

```
Executes a given bash command with optional
timeout. Working directory persists between
commands; shell state (everything else) does not.
The shell environment is initialized from the
user's profile (bash or zsh).

IMPORTANT: This tool is for terminal operations
like git, npm, docker, etc. DO NOT use it for file
operations (reading, writing, editing, searching,
finding files) - use the specialized tools for
this instead.
...
```

Included in every request

Messages: The ongoing conversation

Coding Agent

Included in every request

```
what is the problem with the code in my
Program.cs file?

If you want to read a file, respond with
"Readfile: filename"
```

```
Readfile: Program.cs
```

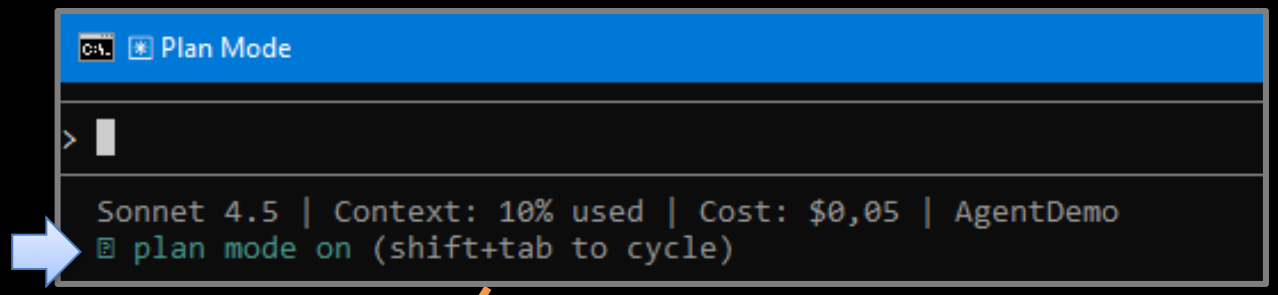
```
Program.cs:
Console.WriteLine("Hello, World!);
```

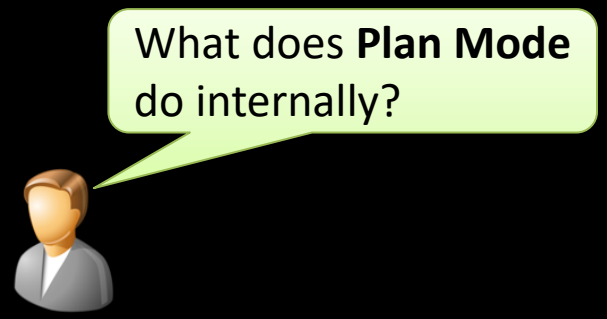
```
The problem is a syntax error caused by
an unterminated string literal.
```

```
Thanks! You are a lifesaver!
```

The latest prompt

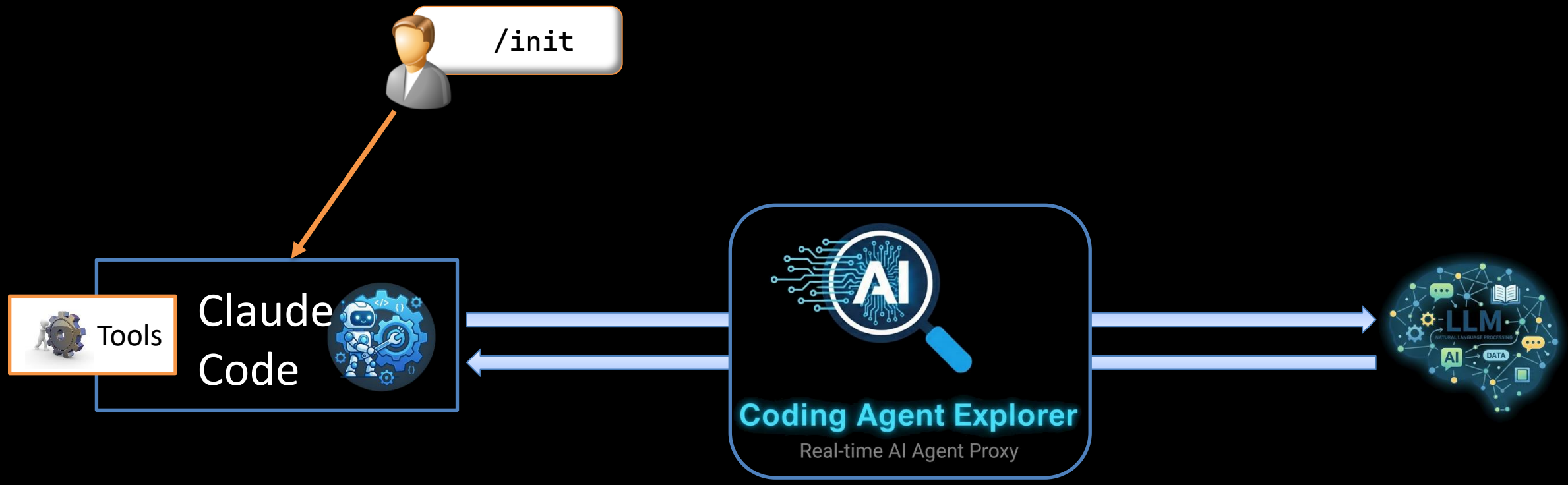
Demo #4 – Exploring the request





Claude Code /init

Demo #5 – Exploring the tools



Building Your Own Agent in C#

An agent is just a loop:

```
var agent = new Agent();

while (true)
{
    var input = GetUserInput();

    var response = agent.SendMessage(input);

    Print(response);
}
```

What about the message history (context)?

The context is just a collection

```
var context = new ContextCollection();

while (true)
{
    var input = GetUserInput();

    context.AddUserPrompt(input);

    var response = agent.SendMessage(context);

    context.AddAgentResponse(response);

    Print(response);
}
```

What about tool support?

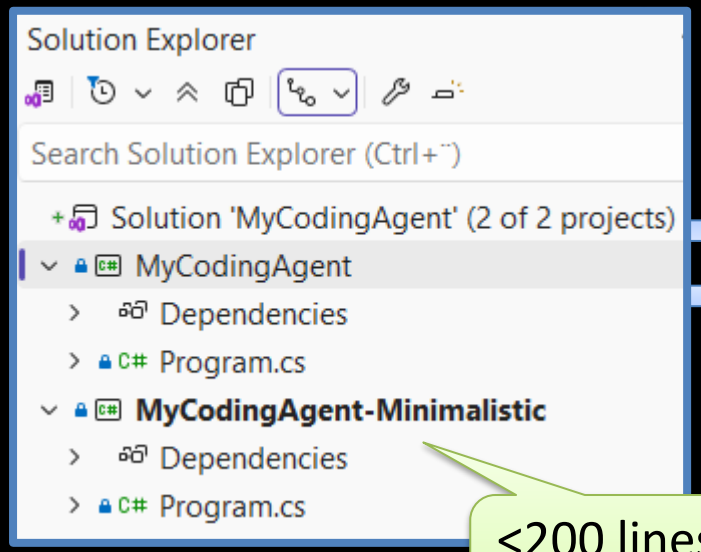
Adding tool support is easy:

```
while (true)
{
    var input = GetUserInput();

    context.AddUserPrompt(input);

    var response = agent.SendMessage(context);

    // Tool chain: read → edit → test
    while (agent.WantsToUseTool(response))
    {
        // Executes tool + appends tool result to context
        response = agent.ProcessToolCall(context, response);
        response = agent.SendMessage(context);
    }

    context.AddAgentResponse(response);
    Print(response);
}
```

Demo #6 – MyCodingAgent.NET

```
Create a simple fizz buzz application in C#, with
program.cs and project file. Do use .NET 10.
```

My Coding Agent .NET



<200 lines of code!

The end!

Do you want this presentation at your work?

Presentation and code
https://github.com/tndataab/PublicBlogContent
https://github.com/tndata/CodingAgentExplorer

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tore@tn-data.se

Work
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QUESTIONS?

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